

TOPIC 21/60

Supervised Fine-Tuning

Teaching the model to be
an assistant, not just a document.

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THE PROBLEM

A raw "Pre-trained" base model is like a genius who has read every book in the library but has **never talked to a human.**

Prompt: "What is the capital of France?"

Base Model: "...is the capital of France, but what about the history of the Eiffel Tower? How many people visit the Louvre..."

It just predicts more text. It doesn't know it should provide a concise answer.

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WHAT IS SFT?



The "Coaching" Phase

We take the base model and train it further on a small, high-quality dataset of perfect "Question/Answer" pairs.

We show it:

Q: "What is 2+2?"

A: "2+2 is 4."

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THE MECHANISM

The Training Loop

The model learns to mimic the desired output style. If the answers in our dataset are polite and professional, the model becomes polite and professional.

Dataset Curation:

Thousands of prompt/completion pairs. This is expensive, labor-intensive work often done by expert human writers.

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KEY DIFFERENCES

Pre-training vs SFT

Pre-training

Trillions of words. Unsupervised. Learn everything.

Fine-Tuning (SFT)

Thousands of examples. Supervised. Learn to follow orders.

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WHY IT'S NECESSARY

1. Format Adoption

Teaches the model to reply in markdown, code blocks, or tables.

2. Instruction Following

Teaches the model to prioritize "answering the question" over "finishing the sentence".

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THE LIMITATION

It doesn't learn new "facts"

SFT is about style and behavior, not world knowledge. If you want it to know facts not in its original pre-training data, you use RAG, not SFT.

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SUMMARY

BASE MODEL

Raw Intelligence (The "Brain")

AFTER SFT

Instruct Model (The "Assistant")

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NEXT STOP

RLHF

Reinforcement Learning from Human Feedback.

Where we teach the model the "Vibe" of human preferences.

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Supervised Fine-Tuning

CONCEPT UNLOCKED

- 🎯 **Objective:** Follow instructions
- 📖 **Data:** Curated Q&A pairs
- ⚙️ **Goal:** Behavioral refinement

CONNECT WITH ME



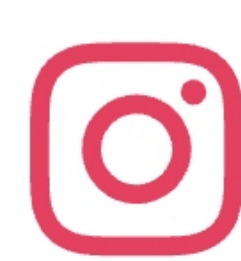
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